A big proof of a small result (suite & fin)

<u>Cor 3.11</u> $F/\mathbb{Q}_p$ finite, $\pi \in F$ a uniformizer. Let $A \subseteq \mathcal{O}_F$ be a lift of F_v. Then any $x \in U_F^{(1)}/U_F^{(k+1)}$ has a unique representation of the form $a U_F^{(k+1)}$ with $a = 1 + a_1\pi + \dots + a_k\pi^k$ & $a_i \in A$.

Note: if $F = \mathbb{Q}_p$, we're writing x as $\sum a_i p^i$; now we're trying to write $x \in F/\mathbb{Q}_p$ as $\sum a_i \pi^i$, up to some k.

Proof of 3.11 existence is by 3.9, uniqueness is easy.

Now we need some calculations:

<u>Lem 3.12</u> $F/\mathbb{Q}_p$ finite and assume $\zeta_p \in F$. Then

- $\dfrac{p}{(\zeta_p - 1)^{p-1}} \equiv -1 \mod (\zeta_p - 1)$

&

- $\dfrac{p}{(\zeta_p - 1)^{p-1}}$ has residue $-1 \in F_v$

<u>Proof</u>: take $f(X) = X^{p-1} + \quad + X + 1 = \dfrac{X^p - 1}{X - 1} = \prod_{i=1}^{p-1} (X - \zeta_p^i)$

We have $f(1) = p$, so $\dfrac{p}{(\zeta_p - 1)^{p-1}} = \dfrac{f(1)}{(\zeta_p - 1)^{p-1}} = \prod_{i=1}^{p-1} \left(\dfrac{1 - \zeta_p^i}{\zeta_p - 1} \right) = (-1)^{p-1} \prod_{i=1}^{p-1} \left(\dfrac{\zeta_p^i - 1}{\zeta_p - 1} \right)$

But $\frac{\zeta_p^i - 1}{\zeta_p - 1} = 1 + \zeta_p + \cdots + \zeta_p^{i-1} = 1 + (1 + (\zeta_p - 1)) + \cdots + (1 + (\zeta_p - 1))^{i-1}$

$\equiv i \mod (\zeta_p - 1)$.

So $\frac{p}{(\zeta_p - 1)^{p-1}} \equiv (-1)^{p-1} (p-1)! \mod (\zeta_p - 1)$

We have $v(\zeta_p - 1) = \frac{1}{p-1}$, so $p \in (\zeta_p - 1)$; and since

$(-1)^{p-1} \equiv 1 \mod p$, $(p-1)! \equiv -1 \mod p$,

We conclude $\frac{p}{(\zeta_p - 1)^{p-1}} \equiv -1 \mod (\zeta_p - 1)$.

<u>Lem 3.13</u> $F/\mathbb{Q}_p(\zeta_p)$ finite, $e = e(F/\mathbb{Q}_p(\zeta_p))$. Then

$$U_F^{(ep+1)} \subseteq \left(U_F^{(e)}\right)^p$$

and if $F = \mathbb{Q}_p(\zeta_p)$, then $U_F^{(p+1)} = \left(U_F^{(1)}\right)^p$.

<u>Proof</u>: $\mathbb{Q}_p(\zeta_p)/\mathbb{Q}_p$ is tot. ram. of deg $p-1$, & $\pi = \zeta_p - 1$ is a uniformizer. Let Π be a unif. for F.

We have $v(p) = 1$, $v(\pi) = \frac{1}{p-1}$, $v(\Pi) = \frac{1}{e(p-1)}$,

so $U_F^{(e)} = 1 + (\pi)$ & $U_F^{(ep+1)} = 1 + (\pi^p \Pi)$.

For $x = 1 + \pi^p a \in U_F^{(ep+1)}$ with $a \in \Pi$. Let

$$f(X) = \pi^{-p}\left[(1+\pi X)^p - (1 + \pi^p a)\right] = X^p - a + \sum_{i=1}^{p-1} \binom{p}{i} \pi^{i-p} X^i$$

We have $v(a) > 0$, $v\left(\binom{p}{i}\pi^{i-p}\right) = v\left(\binom{p}{i}\right) + (i-p)v(\pi)$

$$= 1 + \frac{i-p}{p-1}$$

$$= \frac{p-1+i-p}{p-1} = \frac{i-1}{p-1} > 0 \text{ if } i \geq 2$$

& $v(p\pi^{1-p}) = 0$, $\operatorname{res}(p\pi^{1-p}) = -1$ (by 3.12)

then $\bar{f}(X) = X^p - X$, has simple root in F_v,

& by Hensel f has a root in $\mathcal{O}_F$:

$$\exists\, b \in \mathcal{O}_F \text{ s.t. } \pi^{-p}\Big(\underbrace{(1+\pi b)^p - (1+\pi^p a)}_{=x}\Big) = 0$$

$$\text{so } x = (1+\pi b)^p \in \left(U_F^{(e)}\right)^p.$$

$$\in U_F^{(e)}$$

If $F = \mathbb{Q}_p(\zeta_p)$, then $\pi = \Pi$, & $\forall b \in \mathcal{O}_F$.

$$(1+\pi b)^p \equiv 1 + p\pi b + \pi^p b^p \mod \pi^{p+1}$$

$$\equiv 1 + \pi^p(b^p - b) \mod \pi^{p+1} \quad \left(\text{since } p\pi = \frac{p}{\pi^{p-1}}\pi^p\right)$$

But $F_v = \mathbb{F}_p$ & $b^p \equiv b \mod \pi$; thus $(1+\pi b)^p \equiv 1 \mod \pi^{p+1}$

that is, $U_F^{(1),p} \subseteq U_F^{(p+1)}$.

The next results is not needed for our goal today, but is useful later.

<u>Lem 3.14</u> $F/\mathbb{Q}_p(\zeta_p)$ finite, $\pi \in F$ unif., $e = e(F/\mathbb{Q}_p(\zeta_p))$, $f = f(F/\mathbb{Q}_p(\zeta_p))$

Let $\{b_1, \dots b_f\} \subseteq \mathcal{O}_F$ be a lift of an $\mathbb{F}_p$-basis of F_v

Then $\{\pi\} \cup \{1 + b_i \pi^k\}_{\substack{1 \le k \le ep \\ 1 \le i \le f}}$ generates $F^\times$ modulo $F^{\times p}$.

If $F = \mathbb{Q}_p(\zeta_p)$, then it is even a basis.

<u>Proof sketch</u> We have $F^\times / F^{\times p} \cong \pi^{\mathbb{Z}} / \pi^{p\mathbb{Z}} \times U_F^{(1)} / \left(U_F^{(1)}\right)^p$

π generates $\pi^{\mathbb{Z}}/\pi^{p\mathbb{Z}}$. We then prove that

$\{1 + b_i \pi^k\}_{\substack{\ell \le k \le ep \\ 1 \le i \le f}}$ generates $U_F^{(\ell)}$ mod $U_F^{(ep+1)}$

for any ℓ; proof by induction on ℓ (from $\ell = ep+1$, decreasing). For the case $F = \mathbb{Q}_p(\zeta_p)$, it's a basis by a counting argument.

We can finally study $F^\times / F^{\times q}$ for any q.

<u>Lem 3.15</u> (Degree Lemma) $F/\mathbb{Q}_p$ finite of deg n, $\zeta_q \in F$, q prime.

Then $F^\times / F^{\times q} = \begin{cases} 2 & q \neq p \\ n+2 & q = p \end{cases}$

Proof Recall that $F^\times \cong \pi^{\mathbb{Z}} \times \mu'_F \times U_F^{(1)}$.

If $q \neq p$, then any $x \in U_F^{(1)}$ has a q^{th}-root by Hensel, (because $x \in U_F^{(1)}$ means $\mathrm{res}(x) = 1$)

so $U_F^{(1)} / (U_F^{(1)})^q \cong \{1\}$.

Lem 3.2 $\Rightarrow \mu'_F = \mu_{p^f - 1}$ with $f = f(F/\mathbb{Q}_p)$. (actually, this was only done for $F = \mathbb{Q}_p$, but can be obtained from 3.4)

$\zeta_q \in \mu'_F$ by assumption, so $q \mid (p^f - 1)$ & $\mu'_F / {\mu'_F}^q \cong C_q$.

Since $\pi^{\mathbb{Z}} / \pi^{q\mathbb{Z}} \cong C_q$, we get $F^\times / F^{\times q} \cong C_q \times C_q$.

If $q = p$, we have again $\pi^{\mathbb{Z}} / \pi^{p\mathbb{Z}} \cong C_p$, & we now have $\mu'_F / {\mu'_F}^p \cong \{1\}$. Remains to study $U_F^{(1)} / (U_F^{(1)})^p$.

we have $U_F^{(1)} / (U_F^{(1)})^p \cong \left(U_F^{(1)} / U^{(ep+1)} \right) \Big/ \left(U_F^{(1)p} / U_F^{(ep+1)} \right)$ since by 3.13 $(U_F^{(1)})^p \subseteq U_F^{(ep+1)}$

with $e = e(F/\mathbb{Q}_p(\zeta_p))$. So $n = e(p-1)f$. By 3.11:

$$\left| U_F^{(1)} / U_F^{(ep+1)} \right| = |\mathbb{F}_v|^{ep} = (p^f)^{ep} = p^{n + \frac{n}{p-1}}.$$

Claim The Frobenius $x \mapsto x^p$ induces a well-defined morphism

$$\Phi : U_F^{(1)} / U_F^{(e+1)} \longrightarrow (U_F^{(1)})^p / U_F^{(ep+1)}$$ which is surjective

& has kernel of size p.

Proof of claim: take $\pi \in F$ a uniformizer, then $v(\pi^{e(p-1)}) = 1$. ($\&$ $v(p)=1$)

Thus $(1+b\pi^{e+1})^p \equiv 1 \mod \pi^{ep+1}$ $\forall b \in \mathcal{O}_F$,

that is, $(U_F^{(e+1)})^p \subseteq U_F^{(ep+1)}$, so Φ is well defined & surjective. For the kernel, use 3.11 to represent elements of $U_F^{(1)}/U_F^{(e+1)}$ as $1+a_1\pi+\dots+a_e\pi^e$, where $\{a_i\}_i$ is a lift of $\mathbb{F}_v$, & solve $(1+a_1\pi \cdots a_e\pi^e)^p - 1 = 0 \mod \pi^{ep+1}$.

(details skipped).

& finally:

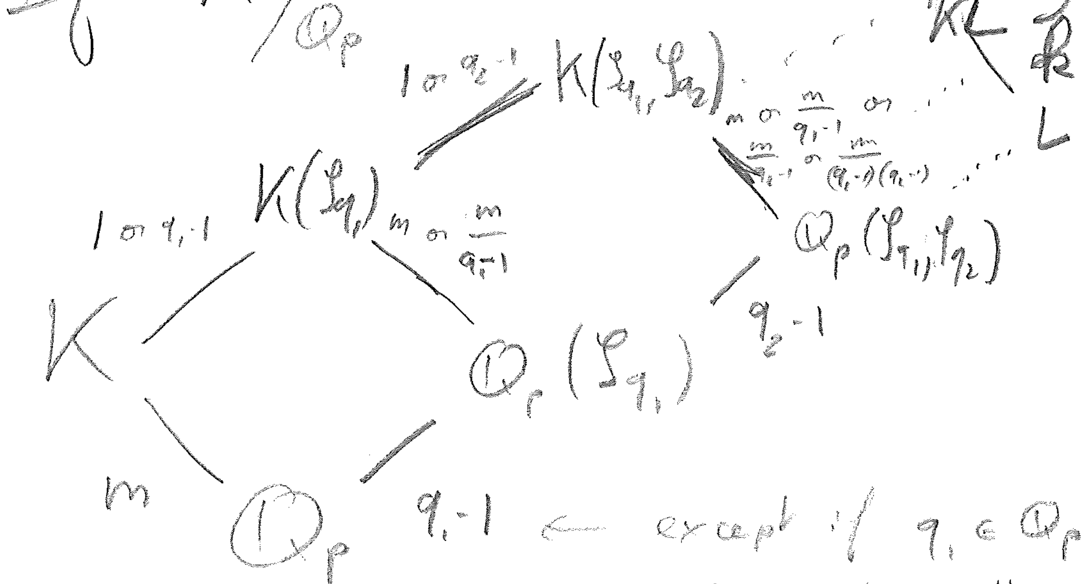
<u>Cor 3.17</u> $G_{\mathbb{Q}_p}$ is small; $\mathbb{Q}_p$ has finitely many distinct extensions of deg n $\forall n$.

<u>Proof</u> assume $\mathbb{Q}_p$ has infinitely many ext. of deg n. Take the normal hull; some distinct ext could have the same normal hull, but only finitely many such. So, $\mathbb{Q}_p$ has infinitely many distinct normal ext of deg $\leq n!$ Pigeonhole $\rightarrow$ $\mathbb{Q}_p$ has infinitely many normal ext of deg m for some m.

Let $L = \mathbb{Q}_p(\zeta_{q_1}, \zeta_{q_2} \cdots \zeta_{q_r})$ where q_i are all prime divisors of m.

Any extension $F(\zeta_q)/F$ is of deg 1 or $q-1$, for F of char not q. here $\mathrm{char}(\mathbb{Q}_p)=0$, of course.

If $K/\mathbb{Q}_p$ is a normal ext of deg m, we have



where $k|m$.

$\leftarrow$ except if $q_1 \in \mathbb{Q}_p$, but then it doesn't matter.

So the infinitely many ~~norml~~ ext of $\mathbb{Q}_p$ of deg m induce infitely many norml ext of L of deg dividing m. Some of them might be trivial, but only finitely many; thus, pigeonhole $\Rightarrow$ L has infinitely many normal ext of degree k for some $k|m$.

They are solvable (in fact, $G_{\mathbb{Q}_p}$ is pro-solvable), so they are a tower of C_q-extensions; since $L \ni \zeta_q$, these are Kummer extensions, of the form $L(\sqrt[q]{x})$ for some $x \in L^\times/L^{\times q}$; there's only finitely many of those by Degree Lemma (3.15).